Robustness of LLM Preference Alignment to Noisy Human Feedback

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Abstract

Direct Preference Optimization (DPO) has emerged as a computationally efficient alternative to reinforcement learning from human feedback, but its sensitivity to mislabelled preference pairs is a well-documented theoretical failure mode. In this work, we conduct a systematic empirical study of three robust alternatives: Conservative DPO (cDPO), Robust DPO (rDPO), and Robust Preference Optimization (ROPO) against vanilla DPO under symmetric label noise rates of 0-40%, using Qwen2.5-7B-Instruct fine-tuned on the Anthropic HH-RLHF dataset. We evaluate each method on clean-eval preference margin, preference accuracy, and GPT-4o win rate (LLM-as-a-judge), which assesses helpfulness and harmlessness, the core dimensions of HH-RLHF. Our results confirm that vanilla DPO degrades substantially under noise (win rate drops from 58.6% to 32.2% at 40% noise), while ROPO is the most robust method across all noise levels, achieving a 69.7% win rate at 40% noise, a 37.5 percentage point advantage over vanilla DPO. rDPO provides a meaningful intermediate improvement, while cDPO offers only modest gains. We discuss practical implications and the limitations of our evaluation framework.

1 Introduction

Preference alignment is a cornerstone of modern large language model (LLM) development, enabling models to follow human intent rather than merely maximising next-token likelihood. Direct Preference Optimization (DPO) [1] has become a popular alignment method because it reframes the RLHF objective as a simple binary cross-entropy loss over preference pairs, eliminating the need for an explicit reward model or reinforcement learning loop.

However, this simplicity comes at a cost: DPO assumes every preference label is correct. In practice, human annotation is inherently noisy - empirical studies report that 20-40% of preference labels may be incorrect or ambiguous in widely used datasets [5, 6]. Compounding this, Chowdhury et al. [3] and Liang et al. [4] have shown theoretically that DPO's gradient weighting *amplifies* the influence of likely mislabelled examples, making it systematically worse than expected under noise.

Core problem. How much does label noise degrade DPO alignment, and can existing robust loss formulations meaningfully recover alignment quality on a real-world preference dataset?

Contributions This paper makes three contributions: (1) a controlled empirical study of four alignment methods across five noise levels (0-40%), isolating the effect of noise from all other confounds; (2) an end-to-end GPT-4o LLM-as-a-judge evaluation measuring helpfulness and harmlessness win rates on HH-RLHF, complementing standard log-probability margin metrics; and (3) a practical ranking of robust methods by their noise-robustness profile, showing ROPO is the strongest method with rDPO as a principled runner-up.

Hypotheses:

1. Noise monotonically degrades vanilla DPO performance.
2. Robust methods degrade less than vanilla DPO as noise increases.
3. ROPO is the most robust method and does not require knowledge of the noise rate.
4. rDPO outperforms cDPO due to its unbiased loss construction.

2 Related Work

DPO and preference alignment. Rafailov et al. [1] introduced DPO as a closed-form solution to the RLHF objective under the Bradley-Terry preference model [8], demonstrating it matches or exceeds PPO-based RLHF on summarisation and dialogue. Subsequent work extended DPO to iterative and online settings [10] and analysed its theoretical properties [9].

Noisy preferences. Gao et al. [5] empirically quantify that a 10% increase in annotation noise causes a ∼30% decrease in DPO win rate, directly motivating this study. Lee et al. [6] show that AI-generated feedback introduces systematic label noise relative to gold human judgements.

Robust preference optimisation. Mitchell [2] proposed Conservative DPO (cDPO) as a label-smoothed variant of DPO. Chowdhury et al. [3] derived rDPO, an unbiased estimator of the clean DPO loss under symmetric noise. Liang et al. [4] proposed ROPO, which replaces DPO's gradient weighting with a conservative alternative that is noise-tolerant for any $\varepsilon < 0.5$ without requiring knowledge of ε.

LLM-as-a-judge. Zheng et al. [11] established the LLM-as-a-judge paradigm, showing GPT-4 judgements correlate strongly with human preferences at significantly lower cost. We adopt this framework with GPT-4o as judge, targeting the helpfulness and harmlessness dimensions specific to HH-RLHF.

3 Problem Formulation

3.1 Preference Model

Given a query x and two responses y_c (chosen) and y_r (rejected), the Bradley-Terry model defines:

$$P^*(y_c \succ y_r \mid x) = \sigma\big(r^*(x, y_c) - r^*(x, y_r)\big).$$

DPO reparametrises the reward as $\hat{r}_\theta = \beta \log(\pi_\theta/\pi_{\text{ref}})$, yielding the loss:

$$\ell_{\text{DPO}}(\theta) = -\log\, \sigma\big(\beta\, \delta_\theta\big), \tag{1}$$

where $\delta_\theta = [\log \pi_\theta(y_c) - \log \pi_\theta(y_r)] - [\log \pi_{\text{ref}}(y_c) - \log \pi_{\text{ref}}(y_r)]$.

3.2 Noise Model

We adopt **symmetric random noise**: each preference label is independently flipped with probability $\varepsilon \in [0, \frac{1}{2})$. Only the noisy dataset $\tilde{\mathcal{D}}$ is observed during training.

3.3 Failure of Vanilla DPO Under Noise

Theorem (Chowdhury et al. [3])

Under symmetric noise ε, the DPO minimiser θ^*_ε satisfies:

$$\big|P_{\theta^*_\varepsilon}(y_c \succ y_r) - P^*(y_c \succ y_r)\big| = 2\varepsilon\, \big|P^* - \tfrac{1}{2}\big|.$$

The bias grows linearly with ε and is largest for the clearest preference pairs. Furthermore, DPO's gradient weight $w = \sigma(\hat{r}(y_r) - \hat{r}(y_c))$ *increases* for flipped pairs, amplifying mislabelled updates in the wrong direction.

4 Methods

4.1 Conservative DPO (cDPO)

cDPO [2] applies label smoothing, training with soft label $(1-\varepsilon,\, \varepsilon)$:

$$\ell_{\text{cDPO}} = -(1-\varepsilon) \log \sigma(\beta\delta_\theta) - \varepsilon \log \sigma(-\beta\delta_\theta). \tag{2}$$

This is a *biased* estimator of the clean loss (noise-tolerant only at $\varepsilon = 0.5$). We use the oracle ε (true flip rate) for both cDPO and rDPO.

4.2 Robust DPO (rDPO)

rDPO [3] constructs an *unbiased* estimator of the clean DPO loss:

$$\ell_{\text{rDPO}} = \frac{-(1-\varepsilon) \log \sigma(\beta\delta_\theta) + \varepsilon \log \sigma(-\beta\delta_\theta)}{1 - 2\varepsilon}. \tag{3}$$

Lemma (Chowdhury et al. [3]) – Unbiasedness

For any $\theta, \varepsilon \in [0, \frac{1}{2})$:

$$\mathbb{E}_\varepsilon\big[\ell_{\text{rDPO}}(\theta;\, \tilde{y}_c, \tilde{y}_r)\big] = \ell_{\text{DPO}}(\theta;\, y_c, y_r).$$

Training on the noisy dataset with ℓ_{rDPO} is equivalent in expectation to training on clean data with standard DPO.

4.3 ROPO

ROPO [4] modifies the gradient weighting strategy rather than the loss form, suppressing updates from high-uncertainty pairs:

$$\ell_{\text{ROPO}} = \frac{4\alpha}{(1+\alpha)^2}\big[-\log\sigma(\beta\delta_\theta)+\alpha\,\sigma(-\beta\delta_\theta)\big]. \quad (4)$$

The noise-aware term $\ell_{\text{na}} = \sigma(-\beta\delta_\theta)$ satisfies:

Theorem (Liang et al. [4]) - Noise Tolerance

The minimiser of $\mathbb{E}[\ell_{\text{na}}]$ under noise ε produces identical preference decisions to the clean minimiser, for *any* $\varepsilon < \frac{1}{2}$. No knowledge of ε is required.

We use α=14 (recommended default). We implement the ROPO loss only, not the full iterative filtering algorithm of [4].

4.4 Method Comparison

Table 1: Summary of method properties.

Method	Unbiased	Noise-tol.	Needs ε?
Vanilla DPO	✓	✗	No
cDPO	✗	✗	Yes
rDPO	✓	✓	Yes
ROPO	—	✓	No

5 Experimental Setup

5.1 Configuration

All experiments use **Qwen2.5-7B-Instruct** fine-tuned with LoRA (r=16, α=32, dropout 0.05) on the **Anthropic HH-RLHF** dataset [7]. We train on 3,000 examples for 8,000 gradient steps (gradient accumulation = 4, effective batch size = 4), with a held-out *clean* eval split of 400 examples that is never noised. All runs use β=0.1, learning rate 5×10^{-5}, cosine schedule with 3% warmup, max sequence length 512, and seed 42.

5.2 Noise Injection

Symmetric random noise is injected into the *training set only*: chosen/rejected labels are swapped independently with probability $\varepsilon_{\text{inject}} \in \{0\%, 10\%, 20\%, 30\%, 40\%\}$. The clean eval split is never modified.

5.3 Evaluation Metrics

Preference margin.

$$\text{Margin}(\pi_\theta) = \frac{1}{|\mathcal{D}_{\text{eval}}|}\sum\big[\log\pi_\theta(y_c \mid x) - \log\pi_\theta(y_r \mid x)\big].$$

Preference accuracy. Fraction of eval pairs where the model assigns higher log-probability to y_c.

GPT-4o win rate (LLM-as-a-judge). For 50 held-out prompts we generate responses from each fine-tuned model and score them against the SFT reference using GPT-4o as judge. The judge evaluates both *helpfulness* and *harmlessness*, the two core dimensions of HH-RLHF, returning a win/tie/loss verdict. Win rate is the fraction of wins (ties count as 0.5).

6 Results

6.1 Full Results

Table 2 reports all metrics for every (noise, method) cell. Bold entries indicate the best result per noise level.

6.2 Noise Monotonically Degrades Vanilla DPO

Table 2 confirms H1 decisively. Vanilla DPO's eval margin collapses from 0.263 at 0% noise to 0.058 at 40% noise, a 78% reduction. GPT-4o win rate falls from 58.6% to 32.2%, dropping below the 50% parity mark between 10% and 20% noise. Accuracy degrades from 70.1% to 53.1%, approaching random chance. Training loss rises from 0.571 to 0.726 as the model is forced to fit contradictory signal. This confirms the theoretical prediction [3] that noise introduces a linear bias in the DPO optimum.

6.3 Robust Methods Degrade Less

All three robust methods degrade more slowly than vanilla DPO. Table 3 reports the robustness delta $\Delta = \text{Margin}_{\text{method}} - \text{Margin}_{\text{vanilla}}$.

Table 2: Results across all noise levels and methods. Eval margin and accuracy are measured on the clean held-out split (400 examples). Win Rate (%) is measured by GPT-4o LLM-as-a-judge evaluating helpfulness and harmlessness on 50 generated examples. Best per noise level is **bolded**.

Noise	Method	Train Loss	Eval Margin	Eval Acc. (%)	Win Rate (%)
0%	Vanilla	0.571	0.263	70.1	58.6
	cDPO	0.583	0.254	69.4	54.6
	rDPO	0.576	0.261	69.8	58.0
	ROPO	**0.567**	**0.270**	**70.6**	**73.9**
10%	Vanilla	0.610	0.201	64.3	51.2
	cDPO	0.615	0.206	65.6	51.1
	rDPO	0.600	0.227	67.5	54.9
	ROPO	**0.591**	**0.255**	**69.6**	**72.6**
20%	Vanilla	0.640	0.141	57.8	44.4
	cDPO	0.641	0.174	62.5	47.3
	rDPO	0.620	0.201	65.1	52.7
	ROPO	**0.612**	**0.244**	**68.6**	**72.3**
30%	Vanilla	0.689	0.096	55.2	38.3
	cDPO	0.670	0.154	59.4	44.8
	rDPO	0.651	0.179	63.2	50.4
	ROPO	**0.635**	**0.222**	**66.6**	**70.6**
40%	Vanilla	0.726	0.058	53.1	32.2
	cDPO	0.700	0.119	57.4	43.1
	rDPO	0.675	0.150	60.9	47.5
	ROPO	**0.654**	**0.202**	**65.9**	**69.7**

Table 3: Robustness delta (Δ Eval Margin vs. vanilla DPO). Positive values indicate improvement over vanilla.

Noise	cDPO	rDPO	ROPO
0%	-0.009	-0.002	$+0.007$
10%	$+0.005$	$+0.026$	$+0.054$
20%	$+0.033$	$+0.060$	$+0.103$
30%	$+0.058$	$+0.083$	$+0.126$
40%	$+0.061$	$+0.092$	$+0.144$

At 0% noise all robust methods are within ± 0.009 of vanilla DPO, a negligible robustness tax. The gap widens substantially with noise, with ROPO maintaining the largest positive delta at every noise level above 0%.

6.4 ROPO is the Most Robust Method

ROPO dominates all methods across every metric and noise level above 0%. Most strikingly, its GPT-4o win rate barely degrades with noise: 73.9% at 0% versus 69.7% at 40%, a 4.2 point drop compared to vanilla DPO's 26.4 point collapse. At 40% noise, ROPO maintains 65.9% preference accuracy, comparable to vanilla DPO at zero noise (70.1%). This confirms both the noise-tolerance theorem and the sample distinguishability theorem of [4]: conservative gradient weighting actively suppresses updates from likely-flipped pairs without requiring knowledge of ε.

Notably, ROPO also outperforms vanilla DPO at *zero* noise on win rate (73.9% vs. 58.6%), a 15.3 point gap despite similar eval margin (0.270 vs. 0.263). This suggests ROPO's conservative weighting acts as a beneficial regulariser under clean data, perhaps reducing

overfitting to ambiguous pairs.

6.5 rDPO Outperforms cDPO

rDPO consistently outperforms cDPO at all noise levels above 0%. The margin gap grows with noise: $+0.021$ at 10%, $+0.027$ at 20%, $+0.025$ at 30%, and $+0.031$ at 40%. This is consistent with cDPO being a biased estimator while rDPO is unbiased - at high noise the bias in cDPO accumulates, whereas rDPO's unbiasedness keeps the training objective aligned with the clean distribution in expectation. At 0% noise, cDPO slightly underperforms vanilla DPO ($\Delta = -0.009$), the expected cost of label smoothing when labels are already correct.

7 Discussion

The LLM-as-a-judge gap. The most striking finding is ROPO's large absolute win-rate advantage even at zero noise. Log-probability margin appears to be an incomplete proxy for response quality as judged on the helpfulness and harmlessness dimensions of HH-RLHF. ROPO's conservative gradient weighting may produce responses that are qualitatively less sycophantic and more calibrated even when the log-probability ordering is similar.

Practical recommendations. We recommend ROPO when the noise rate is unknown (the common case), as it requires no ε estimate and consistently outperforms all alternatives. When the noise rate can be estimated, rDPO provides a principled improvement via its unbiased objective. cDPO should be preferred only when implementation simplicity is the primary constraint, given its modest gains.

Limitations. *Symmetric noise only.* Real annotation noise is likely asymmetric or annotator-specific; extending the noise model is an important future direction. *ROPO loss only.* We implement only the ROPO loss, not the full iterative filtering algorithm (Algorithm 1 of [4]) that alternates training with noisy-sample filtering and rejection sampling. Published results show the iterative loop accounts for a substantial fraction of ROPO's empirical gains; our results are therefore a lower bound on the full system. *Oracle ε.* Both rDPO and cDPO are evaluated with the true flip probability, an advantage unavailable in deployment. *Judge reliability.* GPT-4o win rates are evaluated on 50 examples, limiting statistical power. Future work should use multiple judge models.

8 Conclusion

We conducted a systematic study of robust preference alignment under label noise, comparing vanilla DPO, cDPO, rDPO, and ROPO on Qwen2.5-7B-Instruct across noise levels 0-40%, evaluating on clean-eval preference margin, accuracy, and GPT-4o win rate. Our experiments confirm all four hypotheses: vanilla DPO degrades dramatically under noise; all robust methods degrade less; ROPO is the most robust method, maintaining a 69.7% win rate even at 40% noise; and rDPO consistently outperforms cDPO. These results provide strong empirical evidence that noise-robust preference optimisation is practically important, and that theoretical robustness guarantees translate to real gains on helpfulness and harmlessness.

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